CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B.Tech Examination (IT)

November 2013

IT302.01: Design and Analysis of Algorithm

Date: 22.11.2013,Friday | **Time: 10:00 a.m to 1:00 p.m** | **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(a) Define Time and Space complexity. Explain the three measurement model (Best, Average and Worst) of algorithm analysis with example. 4

(b) An asymptotically fast algorithm running on Slow computer is better than asymptotically slow algorithm is running on fast computer. State True or False and Justify your answer with suitable example. 3

Q-2 14

(a) Write a non-recursive algorithm for computing the nth Fibonacci number. Also compare the time and space efficiency of iterative version with recursive version. Give comment on your answer. 5

(b) If $f(n) = 3n^3 + 5n + 5$ and $g(n) = 5n^2 + 3n + 5$ then check the following equalities: 3

1. $f(n) = \theta(g(n))$ 2. $g(n) = \Omega(f(n))$ 3. $g(n) = O(f(n))$

(c) Solve the following fractional Knapsack Problem using Greedy Algorithm. There are five items whose weights and values are given in following arrays. Weight w[ ] = { 1,2,5,8,9 } Value v[ ] = { 10,6,10, 21, 24 }. The Capacity of Knapsack (W) is 18. 6

OR

Q-2 14

(a) Sort the following list in increasing order of numbers using quick sort. Choose first element as a pivot element. 9, 94, 45, 47, 28, 98, 65, 42, 78, 4, 11, 88, 6 5

(b) Arrange the following terms in increasing order of their running time for large value of n: 3

$n^3, 2^n, \log_2 n, n^2 \log n, n^n, n! - (n-1)!$

(c) What is the importance of asymptotic notations in algorithm analysis? Explain various asymptotic notations used in analysis of algorithm in detail. 6

Q-3 Attempt Any Two. 14

(a) Sort the given numbers using merge sort: 85,24,63,45,17,31,96,50,27,7,90 Also determine asymptotic cost of Divide, Conquer and Combine using recurrence tree method.

(b) Let $P_1, P_2, \ldots, P_n$ be n programs to be stored on a disk. Program P_i requires S_i kilobytes of storage and capacity of the disk is D kilobytes where $D < \sum S_i$ where $1 <= i <= n$. Write a Greedy Algorithm to maximize the number of programs held on the disk.

(c) Solve the following recurrence where n is exactly power of 2.

$T(n) = 1$ if $n=1$

$= 4T(n/2) + n\log n$ otherwise

SECTION-II

Q-4

(a) Consider an undirected unweight graph G .Let a breadth first traversal of G be done starting from a node r. Let d(r, u) and d(r, v) be the lengths of the shortest paths from r to u and v respectively in G. If u is visited before v during the breadth first traversal, which of the following statements is/are correct? **Justify your choice with suitable example.** 2

a. $d(r,u) < d(r,v)$ b. $d(r,u) > d(r,v)$

c. $d(r,u) <= d(r,v)$ d. none of the above

(b) State whether the given statement is true or false with suitable example. 2

If $f(n) = O(g(n))$ and $g(n) = O(f(n))$ then $f(n) = g(n)$.

(c) Is dynamic programming a Top-Down or a Bottom-Up technique? Why? Explain with an example. 3

Q-5 14

(a) Use branch and bound to solve the assignment problem (Minimization) with the following cost matrices. 5

	1	2	3	4
a	94	1	54	68
b	74	10	88	82
c	62	88	8	76
d	11	74	81	21

(b) Give similarities and differences between backtracking and branch-and-bound. Write an algorithm using backtracking to solve "*n queen problem*" and using the algorithm find at least one feasible solution for 5 queen problem. 6

(c) State Master Theorem and Solve following recurrence using Master Theorem. 3

1. $T(n) = 2T(n/2) + n$
2. $T(n) = 4T(n/3) + 1$

OR

Q-5

(a) Solve the following recurrence using recurrence tree method. 5

$T(n) = 3T(n/2) + n$ otherwise

$= 1$ if $n=1$

(b) Let S= {a, b, c, d, e, f, g} be a collection of objects with (weight, benefit) values as follows: a: (2, 2), b: (3, 6), c: (4, 12), d: (5, 20), e: (6, 40). What is an optimal solution to 0/1 knapsack problem using dynamic programming assuming the knapsack can hold objects with total weight 10? 6

(c) Explain Elements of Dynamic Programming Technique. 3

Q-6 Attempt Any Two. **14**

(a) For the following four matrices find the order of parenthesization for the optimal chain multiplication.

A1 = 15 x 5, A2 = 5 x 10, A3 = 10 x 20, A4 = 20 x 25.

(b) Given a finite set of distinct coin types, say 1000,500,100,50,20,10,5,2,1 pcs, and an integer amount C. Each type is available in unlimited quantity. What strategy will you use to find the exact change with minimum no. of coins? Why? Write an algorithm to find the exact change with minimum no. of coins. What is the complexity of algorithm?

(c) Show how depth-first search (DFS) works on the graph of following figure. Assume that the DFS procedure considers the vertices in alphabetical order, and assume that each adjacency list is ordered alphabetically. Show the discovery and finishing times for each vertex, and show the classification of each edge. Source Vertex is Q.